Phase-wise kaam (14-hour SENTINEL GRID build)

Yeh schedule original Winning Plan + aapke master priority list ke hisaab se hai. P0 pehle complete karo (Hour 8 tak), phir P1. Har phase ka exit condition clear rakho.

Phase 0–1 hr: Foundation & contracts

Main objective: Project skeleton, DB models, API contracts, org + asset schema ready.

Exit: Sab interfaces (JSON shapes for warning, asset, exposure, event) agree ho chuke hon.

Phase 1–3 hr: Organisation + Assets + Matcher core

Main objective: Organisation registration, asset CSV upload/validation, synthetic assets, version-aware matcher.

Exit: Organisation ke assets UI mein dikhein aur matcher basic range tests pass kare.

Phase 3–4 hr: Ingestion + Extraction + first match

Main objective: Cached threat feeds load, entity extraction (CVE/IOC/product/version), version matching.

Exit: Ek advisory asset se match ho jaye.

Hour 4 Gate 1 (non-negotiable)

Ek hard-coded/scripted advisory → extract → match → score → warning card UI mein dikhe. Agar nahi hota to feature add karna band karke fix karo.

Phase 4–6 hr: Scorer + Evidence + Warning object

Main objective: Exploitation ladder (L0–L5), transparent scorer (S/E/X/V/I + confidence + rules), evidence/citations, deterministic brief.

Exit: Correct risk score + breakdown + explanation dikhe.

Phase 6–8 hr: Dashboard + Guardrails + Lifecycle + Approval

Main objective: Overview/queue/detail/asset exposure screens, ID verification, citation/injection guardrails, 14-status state machine, approver role, audit log, playbooks.

Exit: Saare P0 acceptance tests pass. (Hour 8 Gate 2)

Phase 8–10 hr: Live demo differentiators (P1 top)

Live inject-advisory + animated pipeline (SSE), KEV escalation + score/band animation.

Exit: Memorable end-to-end flow.

Phase 10–11 hr: Remaining high-value P1

Dependency CSV + exposure graph, what-if patch, logs/investigation (agar time), LLM narrative (citation validation + fallback), polish approval/verification/closure.

Exit: Highest-value P1 features kaam kar rahe hon. (Hour 11 Feature freeze)

Phase 11–12 hr: Freeze + polish

Bug fix, error states, projector contrast, logging, health endpoint, Docker/reset harden.

No new features.

Phase 12–14 hr: Rehearse + package

Clean checkout on second machine (offline), 5 full rehearsals, backup video, slides/README, Q&A drill.

Exit: Repeatable 3-minute demo.

Aapke diye hue roles ke alawa kya kaam hai (gaps)

Aapke current assignment se yeh critical pieces missing hain (Winning Plan ke hisaab se yeh P0 ke liye zaroori hain):

- Backend API layer (FastAPI endpoints, Pydantic models, SSE, seed/reset, health)
- Threat intelligence ingestion + normalisation + deduplication + provenance
- Version-aware matcher + alias table + six exposure statuses (core differentiator)
- Transparent scorer + rules R1–R6 + confidence
- Lifecycle state machine + roles + append-only audit
- Synthetic data prep (40–60 assets, 7 scenario advisories, gold set, playbooks)
- Unit tests (matcher edges, scorer worked examples, state machine)

- Demo infrastructure (offline seed, one-click inject, reset <5s, feature flags)

- Pitch/slides/README/metrics/backup video + Q&A prep

- End-to-end scenario scripting and validation

Research (competitors) already PRD mein cover ho chuka hai – ab build time mein usko minimize karo. Full agentic AI (LangGraph style) plan mein nahi hai; deterministic core + optional LLM (template fallback mandatory) + guardrails prioritise karo. Agentic sirf limited read-only tools + budget + audit ke saath rakhna.

Recommendation: Roles thoda adjust karo taaki vertical slice (Hour 4) aur P0 (Hour 8) safe rahe. Research ko Data/QA person ke under late phase mein shift kar do.

Person-wise Todo lists (aapke roles ke hisaab se, phase-wise mapped)

Person 1 – DB Setup

Focus: Database models, schema, seed/reset, org/asset tables, basic queries.

Hour 0–1

- SQLite (default) + portable schema (PostgreSQL ready)

- Tables: organisation, asset, intel_item, threat, exposure, warning, evidence_link, audit_log, playbook

- Pydantic models + SQLAlchemy

- Seed/reset endpoint skeleton

Hour 1–3

- Organisation registration fields + ID generation

- Asset CSV upload validation (columns, types, criticality, exposure flags)

- Synthetic asset dataset load

- Basic CRUD + association (assets → organisation)

Hour 3–6

- Support for matcher results (exposure status, matched_range, alias_used)

- Warning + evidence_link tables population

- Audit_log append-only design

Hour 6–8

- Lifecycle status fields + transition history support

- Health check queries

- Reset restores deterministic seed <5s

Later (P1)

- Dependency table (source/target/relationship)

- Security log table

- Feature flags for P1

Never cut: Seed/reset, org-scoped assets, exposure status storage.

Person 2 – Research (Competitors + Software)

Focus: Competitive positioning, sources, honest claims. Build time mein isko minimize karo.

Hour 0–2 (limited)

- Quick refresh: MISP, OpenCTI, Wazuh, DefectDojo, SSVC, EPSS/KEV papers (PRD Section 4 & 5 already cover)

- Confirm current public feeds (NVD, CISA KEV, EPSS, CERT-In access constraints)

- List exact claims we will **not** make (no “95% accuracy”, no autonomous action)

Hour 2 onward (shift to Data/QA support)

- Prepare 40–60 synthetic assets CSV

- Write 7 scenario advisories (placeholder CVEs)

- Gold set of 30 items for extraction precision

- Playbooks (versioned)

- Back-test sample (lead time vs KEV) data

- Metrics collection (latency, test pass rates, hallucination catch)

- Slides skeleton + honest caveats

- README sources & limits section

Hour 12–14

- Q&A answers (Section 7.3 style)

- Backup video script support

- Final “what we measured” vs “roadmap” slide

Advice: Pure research ab priority nahi. Data prep + validation + pitch ispe shift karo.

Person 3 – Frontend + RAG

Focus: Complete UX + RAG-style evidence retrieval + narrative rendering.

Hour 0–1

- Vite + React + TS + Tailwind + Framer Motion + Recharts + React Flow

- Design tokens (mission-control dark, teal/amber)

- Layout shell + routing

Hour 1–4

- Organisation registration + asset upload screens

- Asset inventory dashboard

- Overview + warning queue (mock → real API)

- Three-click rule: Queue → Warning → Evidence

Hour 4–8

- Warning detail: risk gauge, dimension bars (S/E/X/V/I), rules fired, evidence panel with citations

- Asset exposure view

- Source health strip + synthetic badge

- Empty/error/degraded states

- Deterministic brief renderer (RAG = retrieve evidence → template)

- Guardrail UI (shield for blocked ID/injection)

Hour 8–11 (P1)

- Live pipeline stepper (SSE events)

- KEV score/band animation

- Exposure/dependency graph (React Flow, max ~15 nodes)

- Approval + status timeline

- What-if UI (if ready)

- LLM narrative display (only after citation/ID validation)

Hour 11–14

- Projector contrast, reduced-motion, keyboard shortcuts (1–7 demo panel)

- Final polish + rehearsals

RAG note: Full vector RAG optional. Priority = evidence retrieval + deterministic brief + citation check. LLM only drafts from retrieved evidence.

Person 4 – Agentic AI

Focus: Matching/AI engine + limited agentic pieces. Full autonomous agents mat banao.

Hour 0–3

- Status & rule enums

- Version parser + alias table + matcher core (packaging + rapidfuzz)

- Six exposure statuses

- Unit tests for range edges (before/through/between/unknown/backport)

Hour 3–6

- Exploitation evidence ladder (L0–L5)

- Transparent scorer (weights + rules R1–R6 + confidence separate)

- Worked example tests (82.0 High → 93.4 Critical on KEV)

- Entity extraction (CVE/IOC/product/version) + gold set precision target

Hour 6–8

- Deterministic brief template
- ID verification (local NVD/ATT&CK)
- Citation validation
- Untrusted feed handling + injection flagging
- Brief renderer (LLM off = works)

Hour 8–11 (P1 / limited agentic)

- LLM gateway (provider-agnostic, strict JSON, template fallback always on)
- Citation + ID check before display
- What-if patch simulation logic
- Limited agent (if any): read-only tools, step budget, full audit, no remediation by model
- Injection demo scenario (poisoned report)

Hour 11–14

- Final unit tests
- Technical Q&A prep
- Hotfix only

Important: Scoring, matching, facts = deterministic. AI only narrative + optional. Guardrails mandatory (show on stage).

Extra kaam jo kisi specific role mein nahi aaya (team share kare)

- Docker Compose + make demo + .env.example
- Full E2E scenario script (7 scenarios)
- Feature flags
- Offline profile (LLM off by default)

- Second laptop clean-checkout test

- Backup video recording

- Honest metrics slide (sample size, uncalibrated weights)

Final non-negotiable rules

- Hour 4 vertical slice fail → sab kuch band, fix karo.

- Hour 8 P0 incomplete → P1 cut from bottom.

- Never cut: org-specific matching, version-range correctness, score breakdown + citations, ID verification, human approval, audit, offline seed + reset.

Agar roles adjust karna chahte ho (recommended: Backend/Ingestion ek person, Matching/AI dusra, Frontend teesra, Data/QA/Pitch chautha) to bata dena – main us hisaab se updated todos de dunga. Abhi ke liye upar wala mapping use kar sakte ho.